

RTL Practice Question Set

Level 1: Combinational Logic

1. Design a 2:1 Multiplexer
2. Design a 4:1 Multiplexer
3. Design a 1:4 Demultiplexer
4. Design a 2-bit Comparator
5. Design a Half Adder
6. Design a Full Adder
7. Design a 4-bit Ripple Carry Adder
8. Design a 4-bit Subtractor
9. Design a Priority Encoder (4:2)
10. Design a Decoder (3:8)
11. Design a Parity Generator
12. Design a Binary to Gray Code Converter

Level 2: Sequential Logic

1. Design a D Flip-Flop (synchronous reset)
2. Design a D Flip-Flop (asynchronous reset)
3. Design a T Flip-Flop
4. Design a 4-bit Register with enable
5. Design a 4-bit Up Counter
6. Design a 4-bit Down Counter
7. Design an Up-Down Counter
8. Design a Mod-10 Counter
9. Design a Shift Register
10. Design a Ring Counter
11. Design a Johnson Counter
12. Design a Serial-In Serial-Out Register

Level 3: FSM Design

1. Design a Traffic Light Controller
2. Design a Sequence Detector (101)
3. Design a Sequence Detector (1101)
4. Design a Vending Machine Controller
5. Design an Elevator Controller
6. Design a Password Lock System
7. Design a UART Transmitter FSM
8. Design a Clock Divider using FSM

9. Design an Overlapping Sequence Detector
10. Design a Handshake Protocol FSM
11. Design a Traffic Controller with delays
12. Design a Serial Data Receiver FSM

Level 4: RTL Systems

1. Design an 8-bit ALU
2. Design a Register File
3. Design a Barrel Shifter
4. Design a Multiplier (shift-add)
5. Design a Simple CPU datapath
6. Design a Memory Controller
7. Design a FIFO Buffer
8. Design UART Transmitter & Receiver
9. Design a Mini RISC Processor
10. Design a Pipelined ALU
11. Design a Cache Controller
12. Design an SPI Controller

Level 5: Verification

1. Write testbench for MUX
2. Write testbench for Counter
3. Write testbench for FSM
4. Self-checking testbench for ALU
5. Random stimulus generation
6. Edge case testing
7. Debug counter issues
8. Fix race condition
9. Identify latch inference
10. Analyze timing violations